

# Energy Ledgers for Forced Harmonic ODEs

Kirchhoff power balance and an unknown component with  $X = LRC$

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## Abstract

Multiplying a forced harmonic LRC equation by current turns its voltage balance into a power balance. Opposite entries identify the heat and source equations needed for a complete energy ledger; their remaining terms are left open. We then add the fictional voltage term  $Xq'''$ , with  $X > 0$  and its physical identity unknown. The sign of its power determines whether it supplies or absorbs energy. Exact numerical examples show both cases. Finally, assuming only the coefficient relation  $X = LRC$ , we check the units and compare how component choices strengthen either contribution to the circuit's power equation.

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Latest on GitHub: <https://github.com/hobnilre/physics-ode-energy>

## 1 Kirchhoff's voltage law and the missing accounts

Consider the forced harmonic equation for a series LRC circuit,

$$Lq'' + Rq' + \frac{q}{C} = f(t), \quad L > 0, \quad C > 0, \quad R \geq 0. \quad (1)$$

Here  $q$  is charge,  $L$  is inductance,  $R$  is resistance,  $C$  is capacitance, and  $f(t)$  is the applied voltage. Coefficients are constant and trajectories are smooth. The equation is second order because its highest derivative is  $q''$ ; it is linear and has degree one in that derivative.

Every term in (1) is a voltage. The component laws are  $v_L = Lq''$ ,  $v_R = Rq'$ , and  $v_C = q/C$ . Kirchhoff's voltage law, written with the source first, is

$$f(t) - Lq'' - Rq' - \frac{q}{C} = 0. \quad (2)$$

Once the ODE is solved, we have  $q(t)$  and  $q'(t)$ . The series current is  $i = q'$ . Multiplication by that current turns the voltage law into a *power law*:

$$f(t)i - Lq''i - Rq'i - \frac{qi}{C} = 0. \quad (3)$$

Each term is now voltage times current, measured in watts. The forcing contributes  $+fi$  to this equation; resistor absorption contributes  $-Rq'i = -Ri^2$ .

The power sum is zero. To relate it to constant total energy, include the heat recipient and the forcing source or sink:

$$\begin{cases} 1 : & \dots + Rq'i + \dots = 0, \\ 2 : & f(t)i - Lq''i - Rq'i - qi/C = 0, \\ 3 : & -f(t)i + \dots = 0. \end{cases} \quad (4)$$

The first line belongs to the heat recipient and the third to the forcing source or sink. Their full ODEs are unknown, but one transfer term in each is already determined. The heat entry  $-Rq'i$  in line 2 has the matching  $+Rq'i$  in line 1. The source entry  $+fi$  in line 2 has the matching  $-fi$  in line 3.

This is the energy counterpart of double-entry economic accounting: a transfer debits one account and credits another. If  $fi < 0$ , the forcing source receives energy, and those entries reverse direction.

The cancellation identifies the transfers needed for a complete ledger. A proof that total energy is constant also requires the omitted terms, including the relevant energy rates and physical reservoir laws. We leave those blanks open here; completing them is a separate article. A prescribed  $f(t)$  alone does not specify a source's available energy.

## 2 Does the unknown term $Xq'''$ supply or absorb power?

### 2.1 The fourth line and the direction of transfer

Add one fictional component term, with constant  $X > 0$ :

$$Xq''' + Lq'' + Rq' + \frac{q}{C} = f(t). \quad (5)$$

Its physical identity is unknown. Its proposed voltage contribution is  $v_X = Xq'''$ . The power equation and its matching entries become

$$\begin{cases} 1: & \dots + Rq'i + \dots = 0, \\ 2: & f(t)i - Xq'''i - Lq''i - Rq'i - qi/C = 0, \\ 3: & -f(t)i + \dots = 0, \\ 4: & \dots + Xq'''i + \dots = 0. \end{cases} \quad (6)$$

The fourth line carries the opposite of the new contribution in line 2. Define the two signed powers explicitly:

$$P_X = Xq'''i \quad (\text{received by } X), \quad P_{X \rightarrow 2} = -Xq'''i \quad (\text{contribution to line 2}). \quad (7)$$

**Table 1.** Direction follows from the derivative product because  $X > 0$ .

| Sign of $q'''i$ | $P_X$    | $P_{X \rightarrow 2}$ | Role of $X$               |
|-----------------|----------|-----------------------|---------------------------|
| Negative        | Negative | Positive              | Supplier                  |
| Positive        | Positive | Negative              | Absorber                  |
| Zero            | Zero     | Zero                  | No instantaneous transfer |

For example, positive current together with negative  $q'''$  makes  $X$  a supplier. Positive current together with positive  $q'''$  makes it an absorber. A positive coefficient alone does not select either role.

Energy transfer follows from the signed power integral:

$$W_X(t_0, t_1) = \int_{t_0}^{t_1} Xq'''(t)i(t) dt, \quad W_{X \rightarrow 2} = -W_X. \quad (8)$$

Positive  $W_X$  is net energy received by  $X$ ; negative  $W_X$  is net energy supplied by it. This specifies the required transfer while leaving the remaining terms of line 4 open.

## 2.2 Two numerical examples from exact trajectories

Take the illustrative values  $L = 1 \text{ H}$ ,  $R = 1 \Omega$ ,  $C = 1 \text{ F}$ , and  $X = 1 \text{ H s}$ . Put  $u = t/(1 \text{ s})$ . In each example choose  $q(t)$  and obtain the required  $f(t)$  by substitution into (5).

**Supplier.** Choose

$$q(t) = (1 \text{ C}) \sin u, \quad i(t) = (1 \text{ A}) \cos u, \quad q'''(t) = -(1 \text{ A/s}^2) \cos u. \quad (9)$$

The inductive and capacitive voltages cancel, as do the resistive and  $X$  voltages, so the required forcing is  $f(t) = 0$ . The two powers are

$$P_X = -(1 \text{ W}) \cos^2 u, \quad P_{X \rightarrow 2} = +(1 \text{ W}) \cos^2 u. \quad (10)$$

At  $t = 0$ ,  $X$  contributes  $+1 \text{ W}$  to line 2. Over the exact period  $T = 2\pi \text{ s}$ ,

$$W_X = -(1 \text{ J}) \int_0^{2\pi} \cos^2 u \, du = -\pi \text{ J}. \quad (11)$$

Thus  $X$  supplies  $\pi \text{ J}$  during the cycle. Here  $P_X = -Ri^2$  pointwise, so the required transfer pays for the resistor's absorption. The equation identifies the required debit in the unknown account; it does not specify what physically supplies that energy.

**Absorber.** On  $0 \leq t \leq 1 \text{ s}$ , choose

$$q(t) = (1 \text{ C}) \frac{u^3}{6}, \quad i(t) = (1 \text{ A}) \frac{u^2}{2}, \quad q'''(t) = 1 \text{ A/s}^2. \quad (12)$$

Direct substitution gives

$$f(t) = (1 \text{ V}) \left( 1 + u + \frac{u^2}{2} + \frac{u^3}{6} \right). \quad (13)$$

Now  $P_X = (1 \text{ W}) u^2/2 > 0$  for  $t > 0$ , so  $X$  absorbs power. At  $t = 1 \text{ s}$ , its contribution to line 2 is  $P_{X \rightarrow 2} = -1/2 \text{ W}$ . Its received energy is

$$W_X = (1 \text{ J}) \int_0^1 \frac{u^2}{2} \, du = \frac{1}{6} \text{ J}. \quad (14)$$

The same positive  $X$  therefore supplies energy on the sinusoid and absorbs energy on the polynomial trajectory. Both examples are exact substitutions and evaluated power integrals.

### 3 Choosing $L$ , $R$ and $C$ when $X = LRC$

#### 3.1 Units and what the coefficient determines

Assume only that the numerical value of the coefficient is given by

$$X = LRC > 0. \quad (15)$$

Consequently  $R > 0$  in this part. The identity and mechanism of  $X$  remain unknown. The relation determines its coefficient value, including units.

Since  $\Omega F = s$ , the dimensional check is

$$[X] = [LRC] = H s = \frac{V s^2}{A}, \quad [q'''] = \frac{A}{s^2}, \quad [Xq'''] = V. \quad (16)$$

Thus the new term is a voltage, and its contribution to line 2 is

$$P_{X \rightarrow 2} = -LRC q''' i. \quad (17)$$

#### 3.2 Maximizing the supplying or absorbing contribution

First hold the trajectory  $q(t)$  fixed, including the time scale and amplitude. Then the derivative product  $q''' i$  is fixed at each instant. Equation (17) gives the two design cases directly:

**Table 2.** At fixed trajectory, both magnitudes increase with the product  $LRC$ .

| Objective at an instant  | Required trajectory sign | Magnitude to maximize |
|--------------------------|--------------------------|-----------------------|
| Supply power to line 2   | $q''' i < 0$             | $LRC(-q''' i)$        |
| Absorb power from line 2 | $q''' i > 0$             | $LRC(q''' i)$         |

Increasing any of  $L$ ,  $R$ , or  $C$  strengthens the existing direction of transfer. Positive component choices cannot reverse that direction for the same trajectory. With independent positive bounds on the three components, the largest product uses all three upper bounds. Without such bounds there is no finite maximum. The signed work in (8) scales in the same way when the trajectory and interval are fixed.

For the sinusoidal family  $q = Q \sin(\omega t)$ , where  $Q > 0$  is charge amplitude and  $\omega > 0$  is angular frequency,

$$q''' = -\omega^2 i, \quad P_{X \rightarrow 2} = LRC \omega^2 i^2 \geq 0. \quad (18)$$

Every positive choice of  $L, R, C$  therefore makes  $X$  a supplier on this trajectory, except at current zeros. Absorption requires a trajectory with  $q''' i > 0$ , such as the polynomial in (12); changing positive coefficients alone cannot produce it on the sinusoid.

If “effect” instead means magnitude relative to the resistor’s absorption, then, at  $i \neq 0$ ,

$$\frac{|P_{X \rightarrow 2}|}{R i^2} = LC \left| \frac{q'''}{i} \right|. \quad (19)$$

At fixed trajectory, larger  $L$  or  $C$  increases this ratio. Larger  $R$  increases both powers equally and leaves the ratio unchanged. On a sinusoid the ratio is  $LC \omega^2$ .

### 3.3 Numerical comparison in both roles

Use the two trajectories from Part 2. Compare the supplier at  $t = 0$  and the absorber at  $t = 1$  s. For each component choice, recompute  $f(t)$  from (5) so that the specified trajectory is retained.

**Table 3.** Exact signed contributions  $P_{X \rightarrow 2}$  of the unknown term.

| $(L, R, C)$ in $(H, \Omega, F)$ | $X$ (H s) | Supplier (W) | Absorber (W) |
|---------------------------------|-----------|--------------|--------------|
| (1, 1, 1)                       | 1         | +1           | -1/2         |
| (1, 2, 1)                       | 2         | +2           | -1           |
| (2, 2, 2)                       | 8         | +8           | -4           |

For example, with independent bounds  $1 H \leq L \leq 2 H$ ,  $1 \Omega \leq R \leq 2 \Omega$ , and  $1 F \leq C \leq 2 F$ , the last choice maximizes both directional magnitudes for the specified trajectories. It supplies 8 W at the supplier instant and absorbs 4 W at the absorber instant. Over their respective intervals, the same exact integrals give  $8\pi$  J supplied during the sinusoidal cycle and  $4/3$  J absorbed during the polynomial interval.

These comparisons hold the motion fixed and adjust the forcing accordingly. If  $f(t)$  and the initial conditions are held fixed instead, changing  $L, R, C$  also changes  $q(t)$ ; maximizing the product alone then does not determine the maximum effect. The coefficient relation specifies the power required of  $X$  on a chosen motion. Its physical identity and the remaining terms of line 4 are still open.